THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** reflection	Reflection.
2	**(a)** rotation	Spinning.
3	**(c)** producers	They make food.
4	**(c)** third planet from the Sun	Order of planets.
5	**(b)** convex	Bulges.
6	**(c)** Purnima	Full moon day.
7	**(d)** food chain	Sequence.
8	**(c)** troposphere	Closest to the ground.
9	**(b)** focal length	f.
10	**(a)** Earth rotates	Apparent motion.
11	**(d)** deforestation	Removal of trees.
12	**(b)** rivers	Not salty.
13	**(d)** angle of reflection	Law.
14	**(d)** 24 hours	A day.
15	**(c)** decomposers	Break down dead matter.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Bouncing back of light from a surface (2).	2
17	Rotation of the Earth (2).	2
18	Living and non-living things interacting in an area (2).	2
19	Crust (1); Mantle and core (1).	1+1
20	Image that cannot be obtained on a screen (2).	2

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Virtual (1); Erect (1); Same size (1).	1+1+1
22	Day and night (1); Apparent motion of the Sun (1); Apparent motion of stars (1).	1+1+1
23	Biotic: fish, algae, frog (1); Abiotic: water, light, temperature (1); They interact (1).	1+1+1
24	Troposphere (1); Stratosphere, mesosphere (1); Thermosphere, exosphere (1).	1+1+1
	OR: n = 360/120 - 1 (1); = 3 - 1 (1); = 2 (1).	1+1+1
25	Natural satellite (1); Reflects sunlight (1); Shows phases (1).	1+1+1
	OR: Provide oxygen (1); Provide wood and food (1); Homes for animals (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Statement of law 1 (1); Statement of law 2 (1); Pins and paper (1); Measure angles (1); i = r (1).	1+1+1+1+1
	OR: Rotation: 24 hours (1); Revolution: 365 days (1); Day and night (1); Seasons (1); Leap year (1).	1+1+1+1+1
27	Biotic (1); Abiotic (1); Producers (1); Consumers (1); Decomposers (1).	1+1+1+1+1
	OR: Crust (1); Mantle (1); Outer core (1); Inner core (1); Features (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** the Moon	One Moon.
2	**(b)** concave	Hollow.
3	**(a)** Amavasya	New moon.
4	**(d)** food web	Network.
5	**(c)** stratosphere	Above the troposphere.
6	**(a)** 10 cm	R/2.
7	**(a)** Pole Star	Near the axis.
8	**(a)** afforestation	Adds forests.
9	**(a)** salty	Contains salts.
10	**(c)** perpendicular to the surface	Right angle.
11	**(c)** one year	Around the Sun.
12	**(d)** herbivore	Plant-eater.
13	**(d)** Earth	Known so far.
14	**(c)** converge	Converge.
15	**(c)** 29.5 days	Lunar month.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Time between two successive new moons, about 29.5 days (2).	2
17	Sequence of organisms in which one eats another (2).	2
18	All water on the Earth (2).	2
19	Concave mirror (2).	2
20	Spinning of the Earth on its axis (2).	2

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Diseases (1); Smog (1); Acid rain (1).	1+1+1
22	It diverges rays (1); Image falls on retina (1); Corrects myopia (1).	1+1+1
23	Sundial uses Sun (1); Watch uses a mechanism (1); Sundial does not work at night (1).	1+1+1
24	Source of energy (1); Photosynthesis (1); Water cycle (1).	1+1+1
	OR: Rivers (1); Lakes (1); Groundwater (1).	1+1+1
25	Dressing (1); Periscope (1); Kaleidoscope (1).	1+1+1
	OR: Year (1); Seasons (1); Change in the night sky (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Regular: mirror (1); Diffuse: paper (1); Both obey laws (1); Image only in regular (1); We see objects due to diffuse reflection (1).	1+1+1+1+1
	OR: Morning (1); Noon (1); Evening (1); Direction (1); Length (1).	1+1+1+1+1
27	Plant trees (1); 3 R's (1); Save water (1); Use public transport (1); Protect animals (1).	1+1+1+1+1
	OR: Lithosphere (1); Hydrosphere (1); Atmosphere (1); Biosphere (1); Links (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** producer	Plants.
2	**(a)** ultraviolet rays	UV.
3	**(d)** 16 cm	2f.
4	**(c)** constellation	Pattern.
5	**(c)** water pollution	Pollution.
6	**(a)** glaciers and ice caps	Frozen.
7	**(a)** virtual	Behind the mirror.
8	**(a)** 366 days	Extra day.
9	**(d)** carnivore	Flesh-eater.
10	**(c)** thin and solid	Outer layer.
11	**(a)** diverge	Spread.
12	**(c)** 27 days	Orbit.
13	**(d)** top carnivore	Eaten by none.
14	**(b)** troposphere	Where clouds are.
15	**(d)** 5 cm behind the mirror	Same distance.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Variety of living organisms (2).	2
17	Troposphere (2).	2
18	Angle of incidence equals angle of reflection (2).	2
19	A year of 366 days (2).	2
20	Grass (1); Trees (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Producers make food (1); Consumers eat food (1); Decomposers recycle (1).	1+1+1
22	Nitrogen 78 per cent (1); Oxygen 21 per cent (1); Other gases 1 per cent (1).	1+1+1
23	f = R/2 (1); 40/2 (1); 20 cm (1).	1+1+1
24	Fixed in the northern sky (1); Shows north (1); Lies near the Earth's axis (1).	1+1+1
	OR: Bacteria (1); Fungi (1); Earthworms help in decay (1).	1+1+1
25	Outermost layer (1); Thin and solid (1); Contains rocks and soil (1).	1+1+1
	OR: Two parallel mirrors (1); Each at 45 degrees (1); Light reflects twice (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Virtual (1); Erect (1); Same size (1); Distance equal (1); Laterally inverted (1).	1+1+1+1+1
	OR: New moon (1); Crescent (1); Half moon (1); Full moon (1); Waning (1).	1+1+1+1+1
27	Definition (1); Grass-deer-tiger (1); Trophic levels (1); Energy flow (1); Decomposers (1).	1+1+1+1+1
	OR: Troposphere (1); Stratosphere (1); Mesosphere (1); Thermosphere (1); Exosphere (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.